# Shoaib Shaikh

Senior Software Engineer - New York Metropolitan Area

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## Summary

Senior software engineer and Tech Lead for Kindle advertising, with 7+ years of deep domain expertise across the digital advertising stack – creative, targeting, forecasting, pacing, delivery, and attribution – building high-scale systems for Kindle advertising and Amazon's device and marketplace advertising platforms. Track record of leading ambiguous, cross-team technical direction end-to-end, from emergency migrations to multi-million-dollar cost, revenue, and compliance initiatives. Recent work has focused on Kindle advertising, including its ad server, targeting, pacing, forecasting, and personalization systems, and on supporting partner teams across other device surfaces.

## Experience

**Amazon – Amazon Ads** July 2025 – Present

*Software Development Engineer II (Tech Lead)* *New York, NY, USA*

- Lead Kindle advertising engineering as Tech Lead, owning the technical roadmap end-to-end, directing teams through the legacy ad server migration and deprecation with zero service disruption, and mentoring 10+ engineers and interns, supporting promotions, new offers, and improved performance ratings.
- Led delivery of European Accessibility Act (EAA) compliance across all Kindle ad placements, including a full legacy template refactor and gap-analysis review with accessibility, product, and legal teams.
- Delivered a $2M annualized revenue increase (20%+ delivery improvement) by architecting an emergency migration off a deprecated targeting service under a hard deadline, discovering and shipping a billable ad-matching optimization for Kindle advertisers.
- Led technical design and presented Kindle's integration into Amazon's cross-platform ad-pacing infrastructure at a multi-team design review, resolving engineering and management concerns from partner teams, and ran a cross-team research spike across 5 teams (forecasting, deals, publisher tech, targeting) to determine feasibility of a new Programmatic Guaranteed deal product on Kindle; findings became durable input to another team's org-wide strategy document months later.
- Designed and delivered end-to-end customer-based personalization signal spanning new data pipelines, feature onboarding, and new runtime fetcher across 5 services – consulting with multiple engineering teams and applied scientists – unblocking a company-wide personalization initiative.
- Oversaw execution of a Kindle-wide Java runtime migration, scoping which devices required it, authoring the validation standard applied across the fleet, unblocking a peer engineer's cutover change, and leading JDK upgrades across four additional dependent services to prevent them from blocking the effort, and led a multi-service migration off a hard-deprecating legacy backend across three production services, independently re-architecting the approach mid-flight after discovering a shared-dependency issue, with zero disruption and a 30% latency improvement.
- Fixed two high-impact delivery and reliability failures: diagnosed a silent ad-delivery failure blocking a next-generation e-reader device's launch, with no logs or errors pointing to a cause and no team yet assigned to it, through a multi-day investigation across app, device, and configuration layers, delivering the fix that unblocked the device's launch; and root-caused a SEV-2 incident spanning two regions to a misconfigured internal throttle rule, led cross-team escalation and resolution, and shipped a durable fix to prevent recurrence.
- Reduced infrastructure costs for Kindle advertising by identifying engineering-level reduction opportunities for hardware and infrastructure spend and partnering with fleet owners and leadership on budget decisions, and by leading a migration from RDS to DynamoDB prompted by a forced MySQL 5.7-to-8 upgrade that would have broken backend services if left unaddressed, cutting infrastructure costs 60% along the way.
- Served as a trusted cross-org point of contact for extending forecasting support across 6 partner teams spanning Alexa, Fire Tablet, and Prime Video, leading requirements and implementation for new ad products, and provided ongoing ad campaign performance guidance to advertising sales leadership, investigating under-performing campaigns and publishing troubleshooting guides used directly by account teams.
- Adopted agentic AI tooling into daily engineering practice – driving automated dead-code detection and cleanup across services and accelerating delivery on multiple migrations and onboardings.

**Humber Polytechnic** January 2023 – August 2025

*Adjunct Professor* *Toronto, ON, Canada*

- Instructed core CS courses (Unix System Administration, Web Development, JavaScript, and Python) to 1000+ students, and created learning materials using practical, hands-on projects and real-world examples.

**Amazon – Amazon Ads** October 2021 – July 2025
*Software Development Engineer II* *Toronto, ON, Canada*

- Designed and led delivery of a service to manage device ad-state migrations from design through production verification
- Built a general-purpose asynchronous, multi-threaded enrollment pipeline, increasing device processing throughput 10x; applied it to safely unenroll 12M devices from a legacy ad program, with headroom for larger-scale future migrations
- Led launch of ad forecasting across Alexa, Fire Tablet, and Prime Video, enabling accurate impression prediction
- Refactored privacy-sensitive decade-old legacy systems under tight deadlines to achieve Digital Markets Act (DMA) compliance with zero downtime
- Led a database right-sizing initiative via blue-green deployments, cutting annual cloud costs 80% ($1.2M)
- Onboarded two new e-reader device lines to ads platform on schedule despite significant technical obstacles, reverse-engineering deprecated ad-package generation tooling and resolving on-device memory issues along the way – generating $2M+ in annual revenue
- De-risked production targeting-service migration by designing multi-step shadow-call validation strategy, and authored initial implementation that peer engineer extended to complete migration

**Amazon – Amazon Ads** February 2020 – October 2021
*Software Development Engineer I* *Toronto, ON, Canada*

- Architected and delivered the compatibility layer that decoupled Kindle's decade-old legacy ad-server from device software, enabling faster and safer ad updates
- Expanded advertising services from US-only to global marketplaces, enabling international scale
- Implemented React web components shown to 100K+ daily active users

**The Citizen Lab – University of Toronto** January 2020 – April 2020
*Research Assistant* *Toronto, ON, Canada*

- Developed ETL pipelines to collect, clean, and analyze large-scale online datasets for academic research

**Amazon – Fulfillment by Amazon (FBA)** May 2019 – August 2019
*Software Development Engineer Intern* *Toronto, ON, Canada*

- Reduced API latency from 500ms to 10ms by eliminating AWS Lambda cold starts via S3-backed caching
- Improved storage performance by regionalizing S3 buckets and optimizing integration tests
- Built internal full-stack debugging tool to replace manual log inspection, slashing developer troubleshooting time

**IBM** May 2018 – May 2019
*Software Developer Intern* *Markham, ON, Canada*

- Reduced automated end-to-end test runtime by 90% via a fail-fast testing strategy and improved CI/CD visibility with automated Slack/email reporting tools

## Education

**Georgia Institute of Technology** Atlanta, GA, USA
*M.S. in Computer Science (Specialization in Computing Systems) – Online* *September 2026 – Present*

**University of Toronto** Toronto, ON, Canada
*Honours B.Sc. in Computer Science (Specialist Focus in Artificial Intelligence), Minor in Economics* *September 2019 – April 2022*

## Technical Skills

**Languages**: Java, Python, JavaScript/TypeScript, C/C++, HTML/CSS, Kotlin, PHP, Ruby, SQL
**Cloud**: AWS( Lambda, DynamoDB, S3, SQS, SNS, RDS, EC2, ECS, Kinesis, Bedrock, IAM, VPC, Route53), Azure
**Frameworks/Runtimes**: Node.js, Express, React, Next.js, Spring Boot, Vue, Angular, Shadcn, Tailwind CSS
**AI/ML & Data**: Generative AI (LLMs, Agentic AI), OpenAI/Claude/Gemini/Grok APIs, PyTorch, TensorFlow, pandas, NumPy, skikit-learn
**Tools & Platforms**: Docker, Git, Jenkins, CI/CD, Supabase, Firebase, Stripe, Cursor
**Concepts**: Digital Advertising, Distributed Systems, Microservices, API Design, Operational Excellence, Technical Debt Reduction